

A Distributed Algorithm for State Estimation with Application to Smart Grid

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Abstract—The next generation power network, i.e., smart grid, stresses distributed technologies, such as distributed power generation, monitoring and storage, etc. Thereby a series of distributed state estimation methods are proposed to decentralize the traditional state estimation methods in power systems. In this paper, a novel distributed WLS state estimation algorithm is developed, in which all local monitors installed in subsystems are only assumed to get access to local network structure information. Boundary estimation information with very low dimension is required to exchange between neighbor subsystems, which leads to a low communication burden. It is shown that after a finite number of iterations, every local estimate can reach the same performance as the corresponding estimates obtained via the centralized state estimation method. The IEEE 118-bus system is used as an example to test the validity of this algorithm. And the solution also shows that the distributed estimation scheme is robust to communication packet loss and time delays.

Index Terms—Weighted least squares, distributed state estimation, power system state estimation

I. INTRODUCTION

Power system state estimation as a core application of the on-line energy management system (EMS) has been well developed since it was introduced in early 1970's [1]. The traditional centralized state estimator is typically installed in central control center in order to provide the best estimate of the state based on all the measurements collected by SCADA systems as well as a global power system model, which usually takes minutes to get a snapshot of a normal sized power system [2], [3]. However, the significant growth of demand for electricity has even stressed the importance of the development of renewable and distributed power generation along with decentralized energy storage technologies in the last decade. In this process, the so-called “smart grid” is considered to replace the traditional highly centralized power system at low voltage levels, where the architecture of the future power grid is fully distributed to distributed generation and distributed control [4]. A series of distributed state estimation algorithms (also known as multi-area state estimation) will become more favored in smart grids to mitigate the contradiction between the increasingly urgent requirement of a better real-time visibility of the operating state and the huge amount of data brought by smart meters. A distributed estimation scheme usually distributes a group of local estimators to replace the centralized estimator based on the usually weaker geographical or measurement coupling properties among different subsystems, in order to reduce the problem size and speed up state estimation in large

scale power systems. According to the structure of the partitioned system, the distributed state estimation algorithms can be generally classified into two categories, i.e., hierarchical estimation and decentralized estimation[5].

Hierarchical estimation methods are usually operated by a set of local estimators and a coordinator, e.g., see [6], [7]. At the subsystem level, local estimates are obtained based on local measurements. Then local estimates are coordinated by the coordinator in order to cope with the interactions among subsystems. In [8], [9], [10], [11], [12] Phasor Measurement Units (PMUs) which offer to provide accurate synchronized measurements of the states with high sampling rate are considered in the state estimation decentralizing process. PMU measurements with good precision are usually utilized to improve the boundary measurement redundancy of local subsystems as well as to coordinate the time skew between different subsystems. In the hierarchical structure, directive communications between every subsystem and the coordinator are required.

Decentralized estimation methods remove the central coordinator, in which neighboring subsystems exchange border information with each other. After local estimates are obtained in a similar fashion to the local estimator in the hierarchical methods, proper local information is exchanged among neighboring subsystems following a designed scheme. Then the corresponding latest obtained information motivates the local estimators to update their own local estimates. Repeating the process for a number of iterations, local information of every subsystem can be broadcast globally. In this way the estimation performance of each final local estimate can get close to the centralized estimates. In [13], a distributed state estimation method was proposed by combining recursive quadratic programming with the dual method, in which global convergence could be guaranteed. Then in a subsequent work by the same authors [14], the distributed state estimation method was improved to be more practical. In recent years, the distributed optimization concept was introduced to state estimation in power systems. In [15], the traditional WLS state estimation method in power systems was formulated into the distributed optimization problem. The convergence to the global optimal solution is shown. A fully decentralized and distributed SW algorithm was proposed in [16]. In this algorithm, each local monitor is only assumed to know local knowledge of the network structure, which is ordered to communicate information with certain neighbors depending on different mode: incremental mode or diffusive mode. After a number of iterations, the error between the decentralized computation results and the global optimal solution can be

bounded.

In this paper, a novel fully decentralized linear weighted least squares (WLS) estimation algorithm is proposed. Taking advantages of the physical linkages between different subsystems, i.e., the common tie-line measurements, and considering the sparse property of the measurement matrix, each local monitor is only in charge of state estimation in its own region based on the topological information of local network, local measurements and the received neighbors boundary estimates. This scheme is mentioned in [16] as a finite-memory estimation technique. However, the local estimate could not converge to the optimal solution, but the one with an upper bound with respect to the optimal one. The algorithm proposed in this paper possesses the following five desired features: (The first two are also owned by [16]): 1) Computation resources are fully decentralized and distributed, and only local knowledge of the subsystem structure is known by the corresponding local estimator. 2) Only low dimensional border information is exchanged, which guarantees a light communication load. 3) Local estimates can reach the same values obtained by the centralized estimator after a finite number of iterations. 4) Each local monitor only need to get access to its own state estimate as well as a small amount of border information from its neighbors, which is important to the security issues of different companies/ISOs/RTOs. 5) The proposed algorithm is robust to communication failures and time delays. In addition, the proposed algorithm is general, whose applicability is not limited to the power system. For instance, it can be used for state estimation in any system with the similar topological structure, such as traffic systems, sensor networks, etc.

This paper is organized as follows. Section II introduces the system model and the problem to be studied. The details of the distributed state estimation algorithm are given in Section III. Section IV provides the simulation results based on the IEEE 118-bus system. Section V concludes the paper.

II. SYSTEM MODELING AND PROBLEM STATEMENT

Consider an interconnected system that can be represented by an acyclic graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ (see for example Fig. 1).

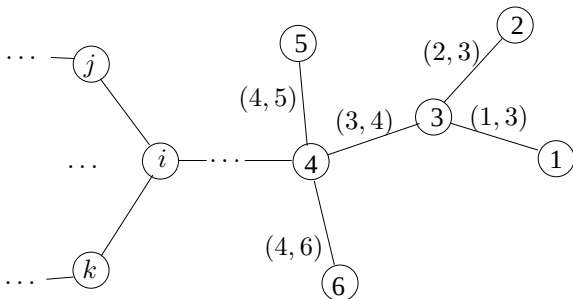


Figure 1. An interconnected system represented by an acyclic graph.

Each node $i \in \mathcal{V} = \{1, 2, \dots, n\}$ represents a subsystem of the interconnected system, whose state is $x_i \in \mathbb{R}^{s_i}$. The edges in the graph indicate the links between different subsystems. Lots of practical systems can be modeled in this way, including power grids, traffic networks and internet. Taking a power grid as an example, a node i refers to a subnetwork in a local

area, while the edges refer to the transmission lines between different subnetworks.

Suppose on each node i in the graph, we have a node measurement, namely,

$$z_i = A_i x_i + w_i, \quad i \in \mathcal{V}, \quad (1)$$

where $z_i \in \mathbb{R}^{q_i}$ is called the measurement on node i , $A_i \in \mathbb{R}^{q_i \times s_i}$ is the measurement matrix, and w_i stands for the measurement noise, which is white Gaussian with zero mean and covariance $\text{cov}(w_i) = R_i$.

Moreover, on each edge $(i, j) \in \mathcal{E}$, we have an edge measurement for which the measurement relates to the states of both nodes associated with this edge. That is,

$$z_{(i,j)} = B_{ij} x_i + B_{ji} x_j + w_{(i,j)}, \quad (2)$$

where $z_{(i,j)} \in \mathbb{R}^{r_{ij}}$ is called the measurement on edge (i, j) , B_{ij} and B_{ji} with appropriate dimensions are the measurement matrices, and $w_{(i,j)}$ is also white Gaussian with zero mean and covariance $\text{cov}(w_{(i,j)}) = R_{(i,j)}$.

For the example of power grid, the node measurement represents the internal measurement for subnetworks where a measurement equipment is placed, and the edge measurement represents the tie-line measurement between two subnetworks.

In the paper, we have the following assumptions.

Assumption 1: The topological structure of graph $\mathcal{G}$ is acyclic.

Assumption 2: A_i is of full column rank for $i \in \mathcal{V}$.

Denote the aggregated state $x = (x_1, \dots, x_n)$ and the aggregated measurement $z = (\dots z_i \dots z_{(i,j)} \dots)$. Then the measurement equation for the graph can be written as

$$z = Hx + \omega, \quad (3)$$

where

$$H = \begin{bmatrix} \vdots \\ H_i \\ \vdots \\ H_{(i,j)} \\ \vdots \end{bmatrix}$$

with $H_i = [0 \ A_i \ 0]$ and $H_{(i,j)} = [0 \ B_{ij} \ 0 \ B_{ji} \ 0]$. In the above 0 indicates a zero matrix with proper dimensions. The covariance of ω is of the block diagonal form, i.e.,

$$R := \text{cov}(\omega) = \text{diag}(\dots \ R_i \ \dots \ R_{(i,j)} \ \dots).$$

Then based on the typical linear measurement equation (3), the traditional weighted least squares (WLS) estimation method can be utilized to calculate the optimal estimation value as

$$\hat{x}^* = \arg \min \left((z - Hx)^T R^{-1} (z - Hx) \right) \quad (4)$$

$$= (H^T R^{-1} H)^{-1} H^T R^{-1} z. \quad (5)$$

The estimation error covariance is given by

$$\begin{aligned} P^* &= E \left((x - \hat{x}^*) (x - \hat{x}^*)^T \right) \\ &= (H^T R^{-1} H)^{-1}. \end{aligned} \quad (6)$$

The above centralized WLS estimation method assumes the complete knowledge of the matrices H and R . In other words, all the measurements z need to be transmitted to a central monitor. Furthermore, Eq. (5) requires the inversion of $H^T R^{-1} H$. Hence a heavy burden of computation and communication is the great challenge for the centralized WLS estimation method applied in very large-scale power systems.

In the paper, we are seeking for light solutions for each node so that it is capable of estimating its own local state, without degrading any estimation performance compared with the centralized estimation scheme. In the proposed distributed estimation algorithm, a number of distributed local monitors are installed to replace the single central monitor. The local monitors need to finish the following two task in estimation: 1) **Local communication**: interchange the boundary estimates with its neighbor nodes; 2) **Local calculation**: run local estimation combining its own local measurements and the boundary estimates received from its neighbor nodes.

The following notions and notations will be used throughout the paper. The set of neighbor nodes of i is defined as $\mathcal{N}_i$. Denote $\mathcal{N}_i/j$ the set of all the neighbor nodes of node i except j . The set of edges linking i and $j \in \mathcal{N}_i$, is defined as $\mathcal{M}_i$.

Now we are ready to introduce the problem formally.

Distributed WLS Problem. Design a local estimation scheme for each node i based on its own node measurement (1), its directly related edge measurement (2), and communication information from its neighbors, to find the unbiased estimates $\hat{x}_i, i \in \mathcal{V}$, minimizing the global objective function

$$J(\hat{x}) := (z - H\hat{x})^T R^{-1} (z - H\hat{x}) \quad (7)$$

where $\hat{x}$ is the aggregated state of local state estimates $\hat{x}_i$.

The motivation for the problem is clear, as for a distributed network over a large scale regions such as power networks, it is usually impossible or costly to collect all the measurements and then make a centralized estimation.

III. DISTRIBUTED WLS ALGORITHM

In this section, we present a distributed WLS estimation scheme and then show that it offers the optimal solution as the centralized one.

We first present our proposed algorithm formally and next give detailed interpretations.

In the algorithm presented below, we use the form of information filter, for which we define

$$\begin{aligned} \alpha_i(k) &= P_i(k)^{-1} \hat{x}_i(k) \\ Q_i(k) &= P_i(k)^{-1} \end{aligned}$$

for any $i \in \mathcal{V}$ and $k = 0, 1, \dots$, called the information vector and information matrix respectively.

First, each node i receives information transmitted from its neighbors, which is calculated via the estimate and corresponding estimation error covariance: (k indicates the iteration step)

$$\beta_j^i(k) = B_{ji} \hat{x}_j(k) \quad (18)$$

$$\Phi_j^i(k) = B_{ji} P_j(k) B_{ji}^T, \quad (19)$$

Algorithm 1 Distributed WLS algorithm.

For each $i \in \mathcal{V}$,

1) Initialization ($k = 0$).

a) Local estimation:

$$\alpha_i(0) = A_i^T R_i^{-1} z_i \quad (8)$$

$$Q_i(0) = A_i^T R_i^{-1} A_i. \quad (9)$$

b) Information transmitting to neighbor $j \in \mathcal{N}_i$:

$$\beta_j^i(0) = B_{ij} Q_i^{-1}(0) \alpha_i(0) \quad (10)$$

$$\Phi_j^i(0) = B_{ij} Q_i^{-1}(0) B_{ij}^T. \quad (11)$$

2) Iteration ($k = 1, 2, \dots$).

a) Information receiving and update:

For any edge (i, j) with $j \in \mathcal{N}_i$, edge measurement update using the estimation from its neighbor:

$$y_{ji}(k) = z_{(i,j)} - \beta_j^i(k-1); \quad (12)$$

Corresponding noise covariance update:

$$S_{ji}(k) = R_{(i,j)} + \Phi_j^i(k-1). \quad (13)$$

b) Local estimation update:

$$\begin{aligned} \alpha_i(k) &= \alpha_i(0) + \sum_{j \in \mathcal{N}_i} B_{ij}^T S_{ji}^{-1}(k) y_{ji}(k) \\ Q_i(k) &= Q_i(0) + \sum_{j \in \mathcal{N}_i} B_{ij}^T S_{ji}^{-1}(k) B_{ij}. \end{aligned} \quad (14)$$

c) Information transmitting to neighbors $j \in \mathcal{N}_i$:

Compute

$$\begin{aligned} \alpha_j^i(k) &= \alpha_i(k) - B_{ij}^T S_{ji}^{-1}(k) y_{ji}(k) \\ Q_j^i(k) &= Q_i(k) - B_{ij}^T S_{ji}^{-1}(k) B_{ij}, \end{aligned} \quad (15)$$

and then transmit the following information to node j

$$\beta_j^i(k) = B_{ij} [Q_j^i(k)]^{-1} \alpha_j^i(k) \quad (16)$$

$$\Phi_j^i(k) = B_{ij} [Q_j^i(k)]^{-1} B_{ij}^T. \quad (17)$$

and then updates the edge measurement using the received information according to the following formula:

$$z_{(i,j)} - \beta_j^i(k) = B_{ji} x_i + B_{ji}(x_j - \hat{x}_j(k)) + w_{(i,j)}. \quad (20)$$

The left-hand side of (20) is treated as a known measurement and the last two terms in the right-hand side of (20) is treated as the measurement noise with covariance $\Phi_j^i(k) + R_{(i,j)}$. Here we would like to point out that the transmitting information (β_j^i and Φ_j^i) relating to the edge measurement is of very small dimension and this indicates a very light communication load.

Second, each node runs a local estimator based on its own measurement (1) and the updated edge measurements (20) to estimate its own state $\hat{x}_i(k+1)$ and also provides the estimation error covariance $P_i(k+1)$.

Third, update the information going to send to the neighbour nodes. We take the information from node i to j , i.e., $\beta_j^i(k+1)$ and $\Phi_j^i(k+1)$, as an example. The local estimator

on node i is ran again based on (1) and (20) (without using the information from j) to calculate $\hat{x}_i'(k+1)$ and $P_i'(k+1)$. Then

$$\beta_i^j(k+1) = B_{ij}\hat{x}_i'(k+1) \quad (21)$$

$$\Phi_i^j(k+1) = B_{ij}P_i'(k+1)B_{ij}^T \quad (22)$$

will be sent to node j .

Iterations continues until every estimate attains the optimal one. It will be shown that after a finite steps related to the “diameter” of the graph, all the estimates will attain the optimal values and further iterations will result in the same values. As a result, each node can terminate its estimation after a fixed number of steps. However, the node should keep transmitting its estimation to its neighbors so that all the nodes in the graph can obtain the optimal estimates.

In the proposed distributed estimation scheme, the resource being used only includes local calculation and local communication with neighbors. Local calculation occurs on each node by only estimating its own state, while communication is only required between neighbors in transmitting a very light data load. Comparing to the centralized WLS estimation scheme that needs to collect the knowledge on all the nodes via communication and needs a powerful computer to do relatively heavy calculations, the proposed distributed scheme can save considerable computation burden and communication bandwidth and thus is particularly useful for large scale networks.

Next we present our main result, but firstly we introduce several notions from graph theory. The maximum distance of a path between node i and any other node j in the graph $\mathcal{G}$ is defined as the *eccentricity* of node i , denoted as ε_i . The maximum eccentricity is called the graph *diameter*, L . Also denote P_i^* ($i = 1, \dots, n$) the block diagonal sub-matrix in P^* corresponding to the state x_i on node i , i.e.,

$$P_i^* = E \left((x_i - \hat{x}_i^*)(x_i - \hat{x}_i^*)^T \right).$$

We start to analyze the proposed algorithm from the following preliminary result.

Lemma 1. Consider a distributed estimation scheme under Assumption 2. Let $\hat{x}_i$ be the resulting estimate and P_i be the associated estimation error covariance, $i = 1, \dots, n$. If $P_i = P_i^*$ for all $i = 1, \dots, n$, then $(\hat{x}_1, \dots, \hat{x}_n)$ equals to $\hat{x}^*$ and minimizes $J(\hat{x})$.

Proof: Assuming the measurement noise has zero mean and diagonal covariance, it is known that if the weighted matrix is chosen as R^{-1} , the WLS estimation $\hat{x}^*$ is also the best linear unbiased estimation that minimizes the mean squared estimation error [17], i.e.,

$$\begin{aligned} \hat{x}^* &= \arg \min \left(E \left(\|x - \hat{x}\|^2 \right) \right) \\ &= \arg \min \left(\text{Tr} \left(E \left((x - \hat{x})(x - \hat{x})^T \right) \right) \right) \end{aligned}$$

For a distributed estimation scheme generating an estimation $\hat{x}_i$ and the associated estimation error covariance P_i ($i = 1, \dots, n$), if $P_i = P_i^*$ for all $i = 1, \dots, n$, then we know

that $\text{Tr}(P^*) = \sum_{i=1}^n \text{Tr}(P_i)$, which means

$$E(\|x - \hat{x}\|^2) = E(\|x - \hat{x}^*\|^2).$$

Note that $E(\|x - \hat{x}\|^2)$ is a convex function and that Assumption 2 ensures the uniqueness of the global minimum for $J(\hat{x})$, so it follows that $(\hat{x}_1, \dots, \hat{x}_n)$ equals to $\hat{x}^*$ and minimizes $J(\hat{x})$. ■

Ensured by Lemma 1, we only need to show $P_i = P_i^*$, $i \in \mathcal{V}$ in the remaining of the paper for a distributed solution.

Theorem 2. Consider an acyclic graph $\mathcal{G}$ and the distributed WLS estimation algorithm. For each node i ($i = 1, \dots, n$), if $k = \varepsilon_i$, then

$$P_i(k+l) = P_i^*, \text{ for all } l \geq 0. \quad (23)$$

Notice that, from (23), the estimate on all nodes converge after $L = \max \{\varepsilon_i, i \in \mathcal{V}\}$ steps. The proof of the theorem requires the following lemmas.

For an acyclic graph $\mathcal{G}$, when we cut off an edge (i, j) , the graph $\mathcal{G}$ becomes two separated subgraphs, which do not connect each other. Denote the subgraph containing node i as $\mathcal{G}_{(i,j)}$. An example is given in Fig. 2.

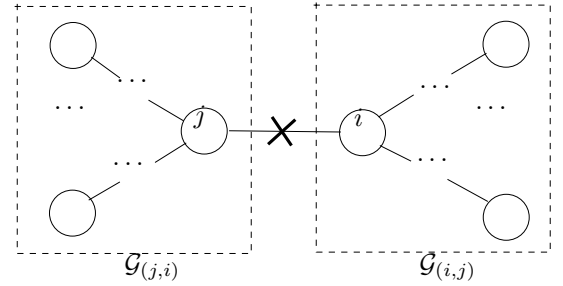


Figure 2. Cutting off edge (i, j) results in two separated subgraphs.

Lemma 3. $(\alpha_i^j(k), Q_i^j(k))$ in the distributed WLS estimator for an acyclic graph $\mathcal{G}$ in step k is the distributed WLS estimate on node i for the subgraph $\mathcal{G}_{(i,j)}$ in step k .

Proof: From (14) and (15), we can get

$$\begin{aligned} \alpha_i^j(k) &= \alpha_i(0) + \sum_{l \in \mathcal{N}_i/j} B_{il}^T S_{li}^{-1}(k) y_{li}(k) \\ Q_i^j(k) &= Q_i(0) + \sum_{l \in \mathcal{N}_i/j} B_{il}^T S_{li}^{-1}(k) B_{il}, \end{aligned}$$

Comparing with the formula (14), it is certain that $(\alpha_i^j(k), Q_i^j(k))$ is the local estimation update on node i for the induced subgraph $\mathcal{G}_{(i,j)}$ after removing edge (i, j) . ■

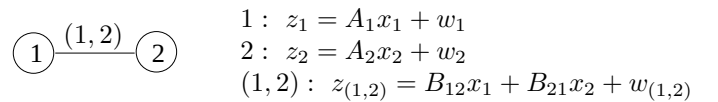


Figure 3. An interconnected system with two nodes.

Lemma 4. Consider an interconnected system of two nodes represented by $\mathcal{G}$ in Fig. 3. Under the distributed WLS scheme,

$$P_i(\varepsilon_i + l) = P_i^*, \quad i = 1, 2, \text{ for all } l \geq 0$$

where $\varepsilon_i = 1$.

Proof: The overall system of the interconnected system can be written as

$$z = Hx + w,$$

where

$$z = \begin{bmatrix} z_1 \\ z_2 \\ z_{(1,2)} \end{bmatrix}, H = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \\ B_{12} & B_{21} \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$w = \begin{bmatrix} w_1 \\ w_2 \\ w_{(1,2)} \end{bmatrix}, \text{ and } R := \text{cov}(w) = \text{diag}(R_1 \ R_2 \ R_{(1,2)}).$$

Consider a centralized WLS estimation scheme. From (??),

$$P^* = (H^T R^{-1} H)^{-1} = \begin{bmatrix} \Gamma_{11} & \Gamma_{12} \\ \Gamma_{12}^T & \Gamma_{22} \end{bmatrix}^{-1}$$

where

$$\begin{aligned} \Gamma_{11} &= A_1^T R_1^{-1} A_1 + B_{12}^T R_{(1,2)}^{-1} B_{12} \\ \Gamma_{12} &= B_{12}^T R_{(1,2)}^{-1} B_{21} \\ \Gamma_{22} &= A_2^T R_2^{-1} A_2 + B_{21}^T R_{(1,2)}^{-1} B_{21}. \end{aligned}$$

According to the matrix inverse lemma, the first diagonal block of P^* is calculated as

$$P_1^* = [A_1^T R_1^{-1} A_1 + B_{12}^T \Omega B_{12}]^{-1} \quad (24)$$

where $\Omega = [R_{(1,2)} + B_{21} P_2(0) B_{21}^T]^{-1}$.

On the other hand, we consider the distributed WLS estimation scheme. Here we look at the estimation on node 1 only as the argument goes the same for the estimation on node 2. It is known that in the initialization step,

$$\begin{aligned} \alpha_i(0) &= A_i^T R_i^{-1} z_i \\ Q_i(0) &= A_i^T R_i^{-1} A_i \quad i = 1, 2, \end{aligned}$$

and the information transmitted from node 2 to node 1 is

$$\begin{aligned} \beta_2^1(0) &= B_{21} Q_2^{-1}(0) \alpha_2(0) \\ \Phi_2^1(0) &= B_{21} Q_2^{-1}(0) B_{21}^T. \end{aligned}$$

In the first step, node 1 updates the edge measurement using the received estimate from node 2 according to the formula in the algorithm as follows:

$$y_{21}(1) = z_{(1,2)} - \beta_2^1(0)$$

and the covariance on the edge measurement error is updated as

$$S_{21}(1) = R_{(1,2)} + \Phi_2^1(0).$$

Thus, according to the formula (14), the local estimation on node 1 is given by

$$\begin{aligned} \alpha_1(1) &= \alpha_1(0) + B_{12} S_{21}^{-1}(1) y_{21}(1) \\ Q_1(1) &= Q_1(0) + B_{12}^T S_{21}^{-1}(1) B_{12}. \end{aligned}$$

Then the estimation error covariance $P_1(1)$ can be calculated as

$$\begin{aligned} P_1(1) &= Q_1(1)^{-1} \\ &= [A_1^T R_1^{-1} A_1 + B_{12}^T S_{21}^{-1}(1) B_{12}]^{-1} \\ &= [A_1^T R_1^{-1} A_1 + B_{12}^T \Omega B_{12}]^{-1}. \end{aligned} \quad (25)$$

Comparing (25) with (24), we know that $P_1(1) = P_1^*$ as $\varepsilon_1 = 1$ for this graph.

Moreover, according to Lemma 3, $(\alpha_2^1(k), Q_2^1(k))$ is the distributed WLS estimate for node 2 for the subgraph $\mathcal{G}_{(2,1)}$, which consists of the single node 2. That is, $\alpha_2^1(k) = \alpha_2(0)$ and $Q_2^1(k) = Q_2(0)$ for $k \geq 1$, which means that the information transmitted to node 1 never changes after step 1. Therefore, the local estimation on node 1 remains the same, i.e.,

$$P_1(1) = P_1(2) = \dots$$

With the same proof, it is also obtained that $P_2(1+l) = P_2^*$ for all $l \geq 0$. ■

Before next lemma, we give a definition of the topological structure of a particular graph.

Definition 5. Radial Graph: An acyclic graph is called a radial graph (shown in Fig. 4), if in this graph, one and only one node's eccentricity equals to 1, and the eccentricities of all the other nodes equal to 2. The node with 1 eccentricity is called the central node, which is indicated by c .

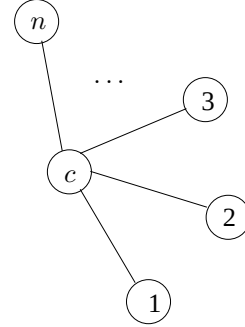


Figure 4. An interconnected system represented by a radial graph.

Lemma 6. Consider an interconnected system represented by a radial graph $\mathcal{G}$ in Fig. 4 with the measurements given by

$$z_i = A_i x_i + w_i, \quad i \in \mathcal{V} \quad (26)$$

$$z_{(c,j)} = B_{cj} x_c - B_{jc} x_j + w_{(c,j)}, \quad j \in \mathcal{N}_c. \quad (27)$$

Under the distributed WLS estimation scheme, we have

$$P_c(\varepsilon_c + l) = P_c^*, \text{ for all } l \geq 0$$

where $\varepsilon_c = 1$.

Proof: Note that the local estimation update on node c is

$$\alpha_c(k) = \alpha_c(0) + \sum_{j \in \mathcal{N}_c} B_{cj}^T S_{jc}^{-1}(k) y_{jc}(k)$$

and

$$Q_c(k) = Q_c(0) + \sum_{j \in \mathcal{N}_c} B_{cj}^T S_{jc}^{-1}(k) B_{cj}.$$

They can be rewritten as

$$\alpha_c(k) = \alpha_c(0) + \begin{bmatrix} B_{c1} \\ B_{c2} \\ \vdots \\ B_{cn} \end{bmatrix}^T \tilde{S} \begin{bmatrix} y_{1c}(0) \\ y_{2c}(0) \\ \vdots \\ y_{nc}(0) \end{bmatrix} \quad (28)$$

and

$$Q_c(k) = Q_c(0) + \begin{bmatrix} B_{c1} \\ B_{c2} \\ \vdots \\ B_{cn} \end{bmatrix}^T \tilde{S} \begin{bmatrix} B_{c1} \\ B_{c2} \\ \vdots \\ B_{cn} \end{bmatrix} \quad (29)$$

where $\tilde{S} = \text{diag}(S_{1c}^{-1}(0), S_{2c}^{-1}(0), \dots, S_{nc}^{-1}(0))$. Eqs. (28) and (29) imply that the radial graph can be treated as a two-node graph by combining nodes $1, \dots, n$ into a single node. Thus, applying Lemma 4, the conclusion follows immediately. ■

Now we come to prove Theorem 2.

Proof of Theorem 2: Consider an acyclic graph $\mathcal{G}$ and a node i in $\mathcal{G}$. Relabel the nodes if necessary so that the graph looks as in Fig. 5 where nodes 1 and 2 are considered to have the maximal distance away from node i .

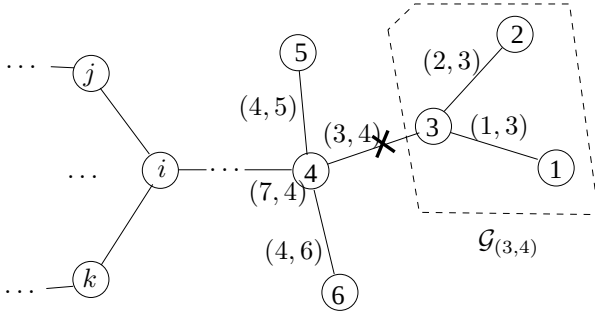


Figure 5. Illustration for the proof of Theorem 2.

In other words, the distance from node 1 and 2 to node i equals to ε_i , while the distance from all other nodes to node i is no more than ε_i .

By cutting off edge $(3, 4)$, it is known that $\mathcal{G}_{(3,4)}$ is a radial graph. Denote ε'_3 the eccentricity of node 3 in the subgraph $\mathcal{G}_{(3,4)}$, which is 1 here. Then it follows from Lemma 6 and Lemma 3 that

$$\begin{aligned} \alpha_3^4(\varepsilon'_3) &= \alpha_3^4(\varepsilon'_3 + 1) = \dots \\ Q_3^4(\varepsilon'_3) &= Q_3^4(\varepsilon'_3 + 1) = \dots \end{aligned}$$

satisfying

$$Q_3^4(\varepsilon'_3) = (P_3^{sub})^{-1} \text{ and } \alpha_3^4(\varepsilon'_3) = (P_3^{sub})^{-1} \hat{x}'_3$$

where $\hat{x}'_3$ and P_3^{sub} are the optimal estimation on node 3 and its corresponding estimation error covariance for the graph $\mathcal{G}_{(3,4)}$ (namely, only using the measurements in $\mathcal{G}_{(3,4)}$).

Next, cut off edge $(4, 7)$ and look at the subgraph $\mathcal{G}_{(3,4)}$. Note that

$$\begin{aligned} \alpha_3^4(1) &= \alpha_3^4(2) = \dots \\ Q_3^4(1) &= Q_3^4(2) = \dots \end{aligned}$$

as we have just shown. Moreover, by the same way, it can be shown that

$$\begin{aligned} \alpha_5^4(0) &= \alpha_5^4(1) = \dots \\ Q_5^4(0) &= Q_5^4(1) = \dots \end{aligned}$$

since the eccentricity of node 5 is 0 in the subgraph $\mathcal{G}_{(5,4)}$. Also, for the same reason we have

$$\begin{aligned} \alpha_6^4(0) &= \alpha_6^4(1) = \dots \\ Q_6^4(0) &= Q_6^4(1) = \dots \end{aligned}$$

Thus according to the formula (15),

$$\begin{aligned} \alpha_4^7(\varepsilon'_4) &= \alpha_4^7(\varepsilon'_4 + 1) = \dots \\ Q_4^7(\varepsilon'_4) &= Q_4^7(\varepsilon'_4 + 1) = \dots \end{aligned}$$

for which $\varepsilon'_4 = 2$ is the eccentricity of node 4 in the subgraph $\mathcal{G}_{(4,7)}$. Moreover, as

$$Q_3^4(\varepsilon'_3) = (P_3^{sub})^{-1} \text{ and } \alpha_3^4(\varepsilon'_3) = (P_3^{sub})^{-1} \hat{x}'_3,$$

by checking the formula (15) at step $k = 2$, it is equivalent to the first-step local estimate on node 4 based on the subgraph $\mathcal{G}_{(4,7)}$ with $\mathcal{G}_{(3,4)}$ being treated as a single node. Thus, applying Lemma 6 and Lemma 3, we know that

$$Q_4^7(\varepsilon'_4) = (P_4^{sub})^{-1} \text{ and } \alpha_4^7(\varepsilon'_4) = (P_4^{sub})^{-1} \hat{x}'_4$$

where $\hat{x}'_4$ and P_4^{sub} are the optimal estimation on node 4 and its corresponding estimation error covariance for the graph $\mathcal{G}_{(4,7)}$.

Since the eccentricity of any node $l \in \mathcal{N}_i$ must be less than or equal to $\varepsilon_i - 1$, repeating this argument eventually leads to the conclusion that for any $l \in \mathcal{N}_i$,

$$\begin{aligned} \alpha_l^i(\varepsilon_i - 1) &= \alpha_l^i(\varepsilon_i) = \dots = (P_l^{sub})^{-1} \hat{x}'_l \\ Q_l^i(\varepsilon_i - 1) &= Q_l^i(\varepsilon_i) = \dots = (P_l^{sub})^{-1} \end{aligned}$$

where $\hat{x}'_l$ and P_l^{sub} are the optimal estimation on node l and its corresponding estimation error covariance for the graph $\mathcal{G}_{(l,i)}$. Thus, from the formula (14), we could get

$$\begin{aligned} \alpha_i(\varepsilon_i) &= \alpha_i(\varepsilon_i + 1) = \dots \\ Q_i(\varepsilon_i) &= Q_i(\varepsilon_i + 1) = \dots \end{aligned}$$

which implies

$$P_i(\varepsilon_i) = P_i(\varepsilon_i + 1) = \dots$$

Furthermore, since $\alpha_l^i(\varepsilon_i - 1) = (P_l^{sub})^{-1} \hat{x}'_l$ and $Q_l^i(\varepsilon_i - 1) = (P_l^{sub})^{-1}$ for any $l \in \mathcal{N}_i$, the local estimate on node i at step $k = \varepsilon_i$ is equivalent to the first-step local estimate on node i for a radial graph with $\mathcal{G}_{(l,i)}$ being treated as a single node. Thus, it follows from Lemma 6 that

$$P_i(\varepsilon_i) = P_i^*.$$

■

IV. EXAMPLES AND SIMULATIONS

The IEEE 118-bus system is utilized to test the proposed algorithm in this paper. The partition of the system is shown in Fig. 6, which can be generally described by the graph in Fig. 7. As mentioned in Section II, each subsystem is seen as a node and the links with tieline measurements are referred as edges. In order to simplify the simulation, only PMU measurements which offer linear measurements are considered. The placement of PMU is optimized by the method proposed in [18]. The system partition is detailed in Table I. We point out that the traditional SCADA measurements also can be utilized in Algorithm 1, after linearization around the operating point [1].

Under Assumption 1 to 3, we run Algorithm 1 on the system described above and also take into account practical situations, i.e., communication packet loss and delays. In the presence of random packet losses or time delay, the behavior

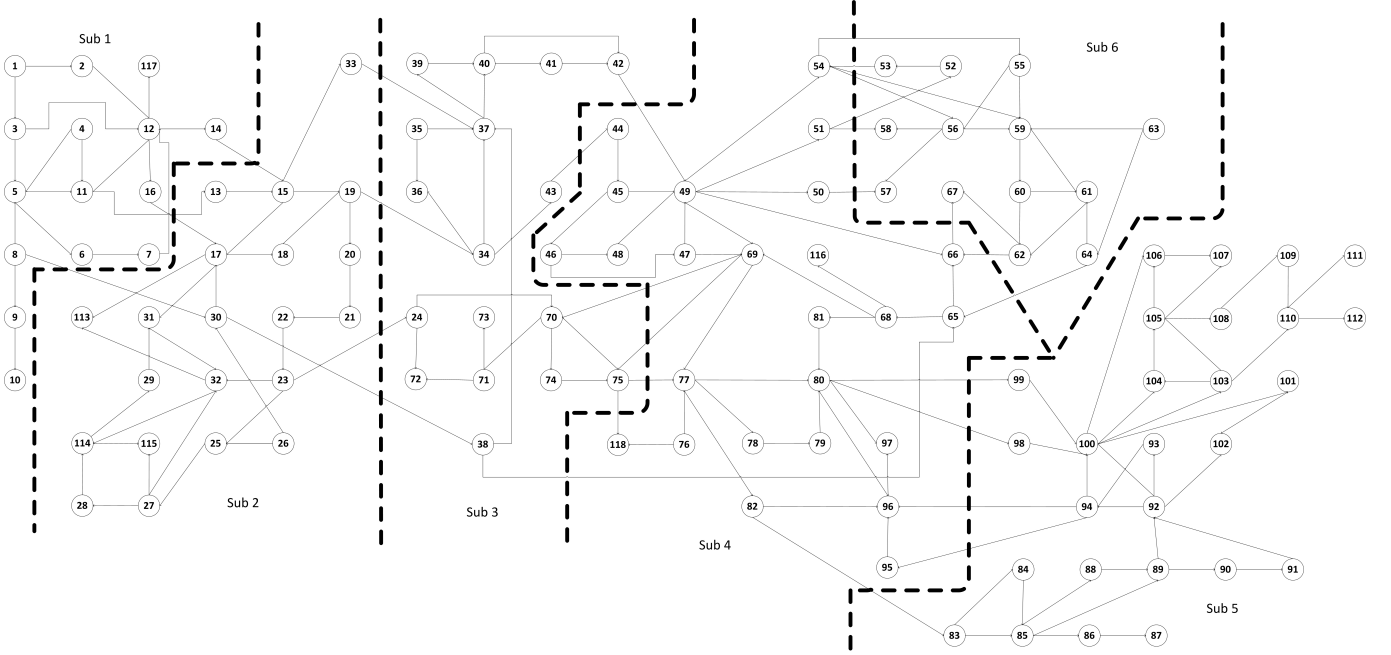


Figure 6. Topological structure of the IEEE 118-bus system.

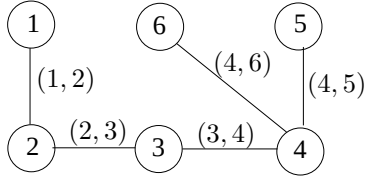


Figure 7. The graph $\mathcal{G}$ describing the partition of the IEEE 118-bus system.

of Algorithm 1, including its convergence time is of random nature. Hence, we use a Monte Carlo simulation to compute, at iteration k , the expected value of the total error between the distributed and centralized estimates, i.e., $\frac{1}{n} \sum_{i=1}^n |\hat{x}_i(k) - \hat{x}_i^*|$, and the expected value of $\sum_{i=1}^n \text{Tr} \{P_i(k)\}$, i.e., sum of the trace of the error covariance of each subsystem. Here we take 1000 Monte Carlo runs.

Fig. 8 and Fig. 9 show the simulation results of packet loss situation, where Bernoulli process is considered to describe packet loss behavior here. Three packet loss rates, i.e., $p = 0.3, 0.15, 0$, are considered. As expected, Fig. 8 and Fig. 9 show that a higher packet loss rate causes a slower convergence speed, but finite-time-convergence is still guaranteed with probability 1 in all cases. The curves corresponding to $p = 0$ in both of Fig. 8 and Fig. 9 show that, as indicated in Theorem 2, the algorithm converges after $k \geq L = 4$ steps.

V. CONCLUSIONS

A novel distributed state estimation algorithm for decentralized local monitors to compute the WLS estimate of the power system state is proposed. The algorithm only requires local topological structure information, local measurements and low dimensional boundary information from neighbors. After a

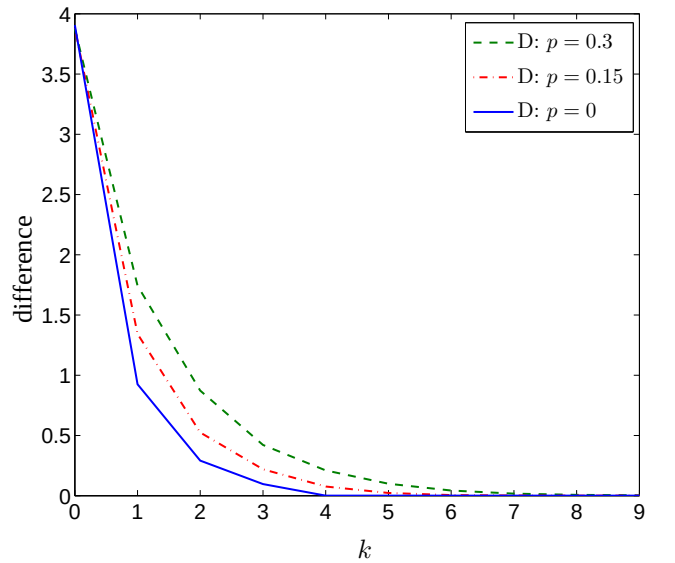


Figure 8. Difference between the distributed (D) and centralized estimates with packet loss.

finite number of iterations, the local estimates can reach the same estimation values obtained via the centralized estimator. The total iteration step equals to the diameter of the graph if no packet loss and time delay occur. The validity of the proposed algorithm can still be assured in some practical operation situations, such as communication packet loss, time delay and subsystem asynchronism. The finite-time-convergence is still guaranteed almost surely.

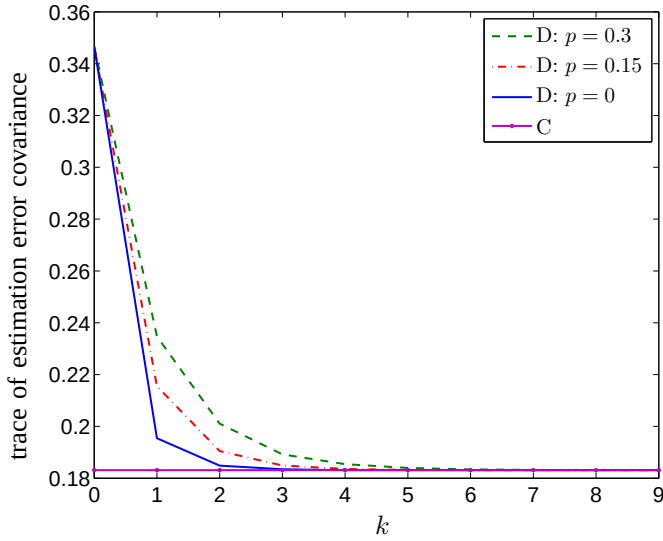


Figure 9. Estimation error covariances of the distributed (D) and centralized (C) estimates with packet loss.

Subsystem	Bus installed PMUs	Bus without PMUs
Sub 1	3 5 9 12	1 2 4 6 7 8 10 11 14 16 117
Sub 2	15 17 21 25 28 114	13 18 19 20 22 23 26 27 29 30 31 32 33 113 115
Sub 3	34 37 40 70 71	24 35 36 38 39 41 42 43 72 73 74 75
Sub 4	45 49 68 76 79 96	44 46 47 48 50 51 54 65 66 69 77 78 80 81 82 95 97 116 118
Sub 5	85 86 92 100 105 110	83 84 87 88 89 90 91 93 94 98 99 101 102 103 104 106 107 108 109 111 112
Sub 6	53 56 62 64	52 55 57 58 59 60 61 63 67

Tie-line	Boundary buses
(1, 2)	14 15 16 17
(2, 3)	19 33 34 37
(3, 4)	42 49 69 70
(4, 5)	53 54 56 62 64 65 66
(4, 6)	94 96

Table I
SYSTEM PARTITION AND MEASUREMENT LOCATIONS

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